

ANDREY KIM

Senior Backend Engineer / Tech Lead • 8+ years
Go, Python, AI Agents • Codeforces Grandmaster (2477)

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EXPERIENCE

Senior Software Engineer / Tech Lead

Tinker Tech 📍 Almaty, Kazakhstan

📅 Feb 2020 – Present

Technically led 3–5 backend engineers across 9+ production systems for banking, fintech, telemedicine, and consulting clients. Drove architecture, CI/CD, and DevOps standards.

- Solo-architected an AI agent platform end-to-end: Python 3.12/FastAPI microservices (agent-api, user, retryer, scheduler), Anthropic Claude API integration with tool use, streaming responses (SSE), custom evaluation harness, agent orchestration, React 19 frontend, Kafka, PostgreSQL 17, Redis Sentinel, Keycloak 26, GitLab CI/CD. Strict tooling: ruff, mypy strict, structlog JSON, async-first (asynccpg, aiokafka, httpx).
- Built Halyk Lifestyle restaurant + payments platform for a major bank (~100 restaurants live across Kazakhstan, averaging ~500 transactions/day, ~3M KZT/day at peak): 18+ Go microservices on K8S/OpenShift, gRPC + Protocol Buffers, Kafka, Redis Sentinel, PostgreSQL, MinIO/S3, Keycloak. Integrations: iiko, rkeeper, osuvox. Designed reusable core for vertical expansion.
- Delivered factoring platforms across two markets (KZ → UZ): Go 1.20–1.23, Gin, GORM, RabbitMQ, PostgreSQL, Redis Sentinel, ArgoCD, Keycloak. Modernized architecture between launches (cmd/config/internal layout, Keycloak SSO).
- Built telemedicine platform with AI assistant: Java 17–21/Spring Boot, OpenAI integration via LangChain4J, Jitsi (video calls), Kafka, PostgreSQL, Keycloak, MinIO, Kubernetes.
- Cut a failing 1-hour Azure CI/CD pipeline to ~15 min by profiling runtime dependencies and stripping 1.5 GB of unused node_modules from the build image (1.5 GB → 200–300 MB).
- Set repo bootstrap, CI/CD, and Docker standards across 9+ projects; introduced an AI-assisted development workflow with explicit output-quality verification.

Coach / Jury, Olympiad Training Center

International IT University 📍 Almaty, Kazakhstan

📅 Apr 2017 – Jan 2020

- Trained the student team that placed 21st at ICPC World Finals, Porto, 2019.
- Mentored the gold medalist of the Republican Olympiad in Informatics (2018).
- Built contest judging infrastructure (Terraform + Ansible on AWS) for IOI-style national olympiads.
- Jury member: International Zhautykov Olympiad (2017–2020, 2025–2026), Republican Olympiad in Informatics (2017–2019).

PROJECTS

LidGuard — macOS Anti-Theft App

Personal Project / Swift 6.2

📅 2025 – Present

- Solo-built native macOS app + privileged helper daemon. Killer feature: motion-based theft detection — accelerometer sampling at ~800 Hz via IOKit/HID directly from the SPU, dual-trigger (15° tilt sustained 400 ms + RMS walking detection).
- Stack: Swift 6.2, SwiftUI, CoreLocation/CoreBluetooth/IOKit/LocalAuthentication/AVFoundation. Telegram bot alerts and remote commands, Bluetooth proximity auto-arm, Touch ID, synthesized siren, fullscreen lock-screen overlay (Sky-Light private API).

ACHIEVEMENTS

- Codeforces Grandmaster — max rating 2477, 124 contests
- Google Software Engineer interview, Warsaw 2019 — passed all 5 technical rounds (process closed without team match)
- Coached ICPC World Finals team — 21st place, Porto 2019
- Alem AI Cup — 1st place • EPAM AI Cup — 1st place • Russian AI Cup — 20th place
- CTF — 2nd place in Kazakhstan (CTFtime, 2021)
- VK Cup Finals — 9th place, 2017 • ACM ICPC Regional — 2nd place, 2015–2016
- Eurasian Olympiad in Informatics — Gold medal, 2012

SKILLS

Languages	Go, Kotlin, Python, C++, Swift, Node.js/TypeScript, Java
AI/LLM	Anthropic Claude API, OpenAI API, LangChain/LangChain4J, agent orchestration, prompt engineering, tool use, streaming (SSE), evals
Infrastructure	Kubernetes/OpenShift, Docker, AWS, Azure, Terraform, Ansible, ArgoCD
Messaging	Kafka, RabbitMQ, gRPC, REST, WebSocket
Databases	PostgreSQL, Redis (Sentinel), MongoDB, MSSQL
Frameworks	FastAPI, Gin, Spring Boot, Django/DRF, React, SwiftUI
Practices	Distributed systems, event-driven architecture, system design, AI-assisted development, CI/CD, Linux, performance profiling
Spoken	English (professional), Russian (native), Kazakh (conversational)

EDUCATION

M.S. Computer Science, International IT University, Almaty (2018–2021) · B.S. Computer Science, IITU (2013–2018)